Logical Equivalence

- **Definition:** Two statement forms are called logically equivalent if, and only if, they have identical truth values for each possible substitution of statements for their statement variables. The logical equivalence of statement forms P and Q is denoted by writing $P \equiv Q$.
- **Testing Whether P and Q Are Logically Equivalent:**
  1. Construct a truth table with one column for the truth values of P and another column for the truth values of Q.
  2. Check each combination of truth values of the statement variables to see whether the truth value of P is the same as the truth value of Q.
     - If in each row the truth value of P is the same as the truth value of Q, then P and Q are logically equivalent.
     - If in some row P has a different truth value from Q, then P and Q are not logically equivalent.

Tautologies and Contradictions

- **Definition:**
  - A *tautology* is a statement form that is always true for all truth values combinations of its variables.
  - A *contradiction* is a statement form that is always false for all truth values combinations of its variables.
  - A *contingency* is a logical proposition that is neither a tautology nor a contradiction.
- **Example:** Show that the statement form $p \vee \sim p$ is a tautology and that the statement form $p \wedge \sim p$ is a contradiction.
- **Solution:**

p	$\sim p$	$p \vee \sim p$	$p \wedge \sim p$
T	F	T	F
F	T	T	F

Column 3 is all T's, so $p \vee \sim p$ is a tautology; While column 4 is all F's, so $p \wedge \sim p$ is a contradiction.

Logical Equivalence Involving Tautologies and Contradictions

Example: If t is a tautology and c is a contradiction, show that $p \wedge t \equiv p$ and $p \wedge c \equiv c$.
Solution:

p	t	$p \wedge t$	c	$p \wedge c$
T	T	T	F	F
F	T	F	F	F

Same truth values, so $p \wedge t \equiv p$ and $p \wedge c \equiv c$.

De Morgan's Laws

- The negation of an **and** statement is logically equivalent to the **or** statement in which each component is negated.

$$\sim (p \wedge q) \equiv \sim p \vee \sim q.$$

- The negation of an **or** statement is logically equivalent to the **and** statement in which each component is negated.

$$\sim (p \vee q) \equiv \sim p \wedge \sim q.$$

Applying De Morgan's Laws

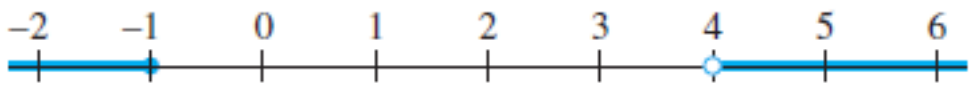
Write negations for each of the following statements:

1. John is 6 feet tall and he weighs at least 200 pounds.
   Solution: John is not 6 feet tall or he weighs less than 200 pounds.
2. The bus was late or Tom's watch was slow.
   Solution: The bus was not late and Tom's watch was not slow.

Inequalities and De Morgan's Laws

Use De Morgan's laws to write the negation of $-1 < x \leq 4$.
Solution: The given statement is equivalent to $-1 < x$ and $x \leq 4$. By De Morgan's laws, the negation is $-1 \geq x$ or $x > 4$, which is equivalent to $-1 \geq x$ or $x > 4$.

Pictorially, if $-1 \geq x$ or $x > 4$, then x lies in the shaded region of the number line, as shown below.



–2 –1 0 1 2 3 4 5 6 ■

Logical Equivalences - Part 1

Given any statement variables p, q, and r, a tautology t and a contradiction c, the following logical equivalences hold:

Law	Logical Equivalence
Commutative laws	$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$
Associative laws	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive laws	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Identity laws	$p \wedge t \equiv p$ $p \vee c \equiv p$
Negation laws	$p \vee \neg p \equiv t$ $p \wedge \neg p \equiv c$
Double negative law	$\neg(\neg p) \equiv p$

Logical Equivalences - Part 2

Law	**Logical Equivalence**
Idempotent laws	$p \wedge p \equiv p$ $p \vee p \equiv p$
Universal bound laws	$p \vee t \equiv t$ $p \wedge c \equiv c$
De Morgan's laws	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$
Absorption laws	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$
Negations of $\top$ and $\bot$	$\neg t \equiv c$ $\neg c \equiv t$

Conditional Statements

Definition: Let p and q be propositions. The conditional statement $p \rightarrow q$ is the proposition "if p, then q." The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the hypothesis (or antecedent or premise) and q is called the conclusion (or consequence).

Conditional Statements

Definition:

Remark: The statement $p \rightarrow q$ is called a conditional statement because $p \rightarrow q$ asserts that q is true on the condition that p holds. A conditional statement is also called an implication.

The Truth Table for the Conditional statement $p \rightarrow q$:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Remark: A conditional statement that is true by virtue of the fact that its hypothesis is false is often called vacuously true or true by default.

Example: The statement "If you show up for work Monday morning, then you will get the job" is vacuously true if you do not show up for work Monday morning.

In general, when the "if" part of an if-then statement is false, the statement as a whole is said to be true, regardless of whether the conclusion is true or false.

Example

If today is Friday, the 2+3=5.
This statement is true by definition of implication since its conclusion is true - The truth value of the hypothesis does not matter.

Example

If today is Friday, then 2+3=6. This statement is true everyday except Friday, Even though 2+3=6 is False.

Order of Operations

In expressions that include $\rightarrow$ as well as other logical operators such as $\wedge$, $\vee$, and $\sim$, the order of operations is that $\rightarrow$ is performed last. Thus, according to the specification of order of operations, $\sim$ is performed first, then $\wedge$ and $\vee$, and finally $\rightarrow$.

Other Terminologies for Conditional Statements

Other terminologies that express conditional statements:

- "p implies q"
- "if p, q"
- "p only if q"
- "p is sufficient for q"
- "a sufficient condition for q is p"
- "q if p"
- "q whenever p"
- "q when p"
- "q is necessary for p"
- "a necessary condition for p is q"
- "q follows from p"
- "q unless $\sim p$"

Representation of Conditional as Disjunction: The conditional "if p then q" is logically equivalent to "not p or q."

$$p \rightarrow q \equiv \sim p \vee q.$$

(Verify this with a truth table)

Example: Rewrite Without Using the Conditional

Example: Rewrite the following statements without using the conditional:

1. If it is cold, he wears a hat.
2. If productivity increases, then wages rise.

Recall that "If p then q" is equivalent to "Not p or q;" that is, $p \to q \equiv \sim p \vee q$.

Hence, we have:

1. It is not cold or he wears a hat.
2. Productivity does not increase or wages rise.

The Negation of a Conditional Statement

The negation of "If p then q" is logically equivalent to "p and not q."

$$\sim (p \to q) \equiv p \wedge \sim q.$$

You can also obtain this result by starting from the logical equivalence $p \to q \equiv \sim p \vee q$. Take the negation of both sides to obtain:

$$\sim (p \to q) \equiv \sim (\sim p \vee q)$$

$$\equiv \sim (\sim p) \wedge (\sim q) \text{ by De Morgan's laws}$$

$$\equiv p \wedge \sim q \text{ by the double negative law.}$$

Example: Write Negations

Example: Write negations for each of the following statements:

1. If my car is in the repair shop, then I cannot get to class.
2. If Sara lives in Athens, then she lives in Greece.

Solution:

1. My car is in the repair shop and I can get to class.
2. Sara lives in Athens and she does not live in Greece.

The Contrapositive of a Conditional Statement

Definition: The contrapositive of a conditional statement of the form "If p then q" is "If $\sim q$ then $\sim p$."
Symbolically, The contrapositive of $p \rightarrow q$ is $\sim q \rightarrow \sim p$.
A conditional statement is logically equivalent to its contrapositive. (Verify)

The Converse and Inverse of a Conditional Statement

Definition: Suppose a conditional statement of the form "If p then q" is given:

- The **converse** is "If q then p."
- The **inverse** is "If $\sim p$ then $\sim q$."

Symbolically,

- The converse of $p \to q$ is $q \to p$.
- The inverse of $p \to q$ is $\sim p \to \sim q$.

Remark:

1. A conditional statement and its converse are not logically equivalent. (Verify)
2. A conditional statement and its inverse are not logically equivalent. (Verify)
3. The converse and the inverse of a conditional statement are logically equivalent to each other. (Verify)

Example: Contrapositive, Converse, and Inverse

Example: What are the contrapositive, the converse, and the inverse of the conditional statement: "The home team wins whenever it is raining?"

Solution: Because "q whenever p" is one of the ways to express the conditional statement $p \rightarrow q$, the original statement can be rewritten as "If it is raining, then the home team wins."

Consequently, the contrapositive of this conditional statement is "If the home team does not win, then it is not raining."

The converse is "If the home team wins, then it is raining."

The inverse is "If it is not raining, then the home team does not win."

The Biconditional Statement

Definition: Let p and q be propositions. The biconditional statement denoted $p \leftrightarrow q$ is the proposition "p if and only if q." The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise.
Biconditional statements are also called bi-implications.
There are some other common ways to express $p \leftrightarrow q$: These include:

- "p is necessary and sufficient for q"
- "if p then q, and conversely"
- "p iff q."

Truth Table for the Biconditional Statement

Truth Table for the Biconditional $p \leftrightarrow q$:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

In order of operations, $\leftrightarrow$ is coequal with $\rightarrow$. As with $\wedge$ and $\vee$, the only way to indicate precedence between them is to use parentheses.

Example: Truth Table Relationship

Example: Truth Table Showing that $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

p	q	$p \rightarrow q$	$q \rightarrow p$	$p \leftrightarrow q$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T
				$\uparrow$	$\uparrow$

$p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$ always have the same truth values, so they are logically equivalent.

Necessary and Sufficient Conditions

Definition: If r and s are statements:

- r is a sufficient condition for s means "if r then s."
- r is a necessary condition for s means "if not r then not s."

In other words, to say "r is a sufficient condition for s" means that the occurrence of r is sufficient to guarantee the occurrence of s. On the other hand, to say "r is a necessary condition for s" means that if r does not occur, then s cannot occur either.

Converting a Sufficient Condition to If-Then Form

Example: Rewrite the following statement in the form "If A then B:"

> *Pia's birth on Kenyan soil is a sufficient condition for her to be a Kenyan citizen.*

Solution: If Pia was born on Kenyan soil, then she is a Kenyan citizen.

Converting a Necessary Condition to If-Then Form

Example: Use the contrapositive to rewrite the following statement in two ways:

> *George's attaining age 35 is a necessary condition for his being president of the United States.*

Solution:

- Version 1: If George has not attained the age of 35, then he cannot be president of the United States.
- Version 2: If George can be president of the United States, then he has attained the age of 35.

Exercise: Proving the Negation of the Biconditional

Exercise: Prove that the negation of the biconditional "p if and only if q" $(\sim (p \leftrightarrow q))$ is equivalent to the exclusive disjunctive form "Either p or q, but not both" $(p \oplus q)$.

Simplifying Statement Forms

Verify the Logical Equivalence:

$$\sim (\sim p \wedge q) \wedge (p \vee q) \equiv p.$$

Solution:

$$\begin{aligned}
&\sim (\sim p \wedge q) \wedge (p \vee q) \\
&= (\sim (\sim p) \vee \sim q) \wedge (p \vee q) \quad \text{by De Morgan's laws} \\
&= (p \vee \sim q) \wedge (p \vee q) \quad \text{by the double negative law} \\
&= p \vee (\sim q \wedge q) \quad \text{by the distributive law} \\
&= p \vee (q \wedge \sim q) \quad \text{by the commutative law for } \wedge \\
&= p \vee c \quad \text{by the negation law} \\
&= p \quad \text{by the identity law.} \qquad \square
\end{aligned}$$

Logical Equivalence Proofs

Claim 1:

$$(p \wedge \sim q) \vee (p \wedge q) \equiv p.$$

Proof:

$$\begin{aligned}(p \wedge \sim q) \vee (p \wedge q) &\equiv p \wedge (\sim q \vee q) \quad \text{by (a)} \\ &\equiv p \wedge (q \vee \sim q) \quad \text{by (b)} \\ &\equiv p \wedge t \quad \text{by (c)} \\ &\equiv p \quad \text{by (d).}\end{aligned}$$

Therefore, $(p \wedge \sim q) \vee (p \wedge q) \equiv p$. $\square$

Claim 2:

$$(p \vee \sim q) \wedge (\sim p \vee \sim q) \equiv \sim q.$$

Proof:

$$\begin{aligned}(p \vee \sim q) \wedge (\sim p \vee \sim q) &\equiv (\sim q \vee p) \wedge (\sim q \vee \sim p) \quad \text{by (a)} \\ &\equiv \sim q \vee (p \wedge \sim p) \quad \text{by (b)} \\ &\equiv \sim q \vee c \quad \text{by (c)} \\ &\equiv \sim q \quad \text{by (d).}\end{aligned}$$

Exercise

Claim 4:

$$p \to (q \to p) \equiv t$$

Proof: Left as Exercise.

Translating English Statements to Logical Expressions

Example 1: You can access the internet from campus only if you are a computer major or you are not a freshman.
Let p be "You can access the internet from campus," q be "You are a computer major," and r be "You are a freshman."

$$(q \vee \neg r) \rightarrow p$$

Example 2: You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old.
Let p be "You can ride the roller coaster," q be "You are under 4 feet tall," and r be "You are older than 16 years old."

$$(q \wedge \neg r) \rightarrow \neg p$$

Example 3: Consider the following propositions:

$$p : \text{I will eat my socks,}$$
$$q : \text{It is raining,}$$
$$r : \text{I will go shopping.}$$

If it is raining, I will eat my socks but I will not go shopping.

$$q \rightarrow (p \wedge \neg r)$$

Example 4: It is either raining or I will go shopping but I will eat my socks if it is raining.

$$(q \vee r) \wedge (q \rightarrow p)$$

Example 5: That it is raining is a necessary and sufficient condition for me not to go shopping.

$$q \leftrightarrow \neg r$$

Translating English Statements to Logical Expressions (cont.)

Example 6: Let p and q be the following propositions:

$$p : \text{It is below freezing,}$$

$$q : \text{It is snowing.}$$

Propositions:

1. It is not below freezing and snowing: $\neg(p \wedge q)$
2. It is below freezing but not snowing: $p \wedge \neg q$
3. It is not below freezing and not snowing: $\neg p \wedge \neg q$
4. It is either snowing or below freezing: $q \vee p$
5. It is below freezing but it is snowing: $p \wedge q$
6. Neither is it below freezing nor is it snowing: $\neg p \wedge \neg q$
7. If it is either below freezing or it is snowing but it is not snowing if it is below freezing: $(q \vee p) \wedge (\neg q \rightarrow p)$
8. If it is below freezing, it is also snowing: $p \rightarrow q$